

MC: Monte-Carlo Learning

for $s_1, a_1, r_1, s_2, a_2, r_2, \dots$ under π

def: Return $G_t = r_t + \gamma r_{t+1} + \gamma^2 r_{t+2} + \dots + \gamma^n r_{t+n}$

def: Value Function: $V(s) = \mathbb{E}[G_t | S_t = s]$ "Terminase"

Empirical Mean Return

- Model-Free: no knowledge of MDP transition/reward
- Monte Learning from Complete episode
- Goal: learn $V_\pi(s)$ ($\neq$ not $V(s)$)

what we want: $V(s)$ converge to $V_\pi(s)$ as $N(s) \rightarrow \infty$.

Actually will. Since Law of Large Number

Every-visit MC: average return for every-time s is visited in an episode.
First-visit MC: average return for first-time s is visited in an episode.

everytime MC

For Everytime s is visited in an episode:

- $N(s) = N(s) + 1$
 $\Rightarrow V(s) = S(s) / N(s)$
- $G(s) = S(s) + G_t$

MC: Policy Evaluation

Input: a policy π to be evaluated
Initialize: in Practice: $V(s) = 0$,
 $V(s) \in \mathbb{R}$, arbitrarily, for all $s \in \mathcal{S}$
 $Returns(s) \leftarrow$ an empty list, for all $s \in \mathcal{S}$
Loop forever (for each episode):
Generate an episode following π : $S_0, A_0, R_1, S_1, A_1, R_2, \dots, S_{T-1}, A_{T-1}, R_T$
 $G \leftarrow 0$
Loop for each step of episode, $t = T-1, T-2, \dots, 0$:
 $G \leftarrow \gamma G + R_{t+1}$
Unless S_t appears in $S_0, S_1, \dots, S_{t-1}$:
Append G to $Returns(S_t)$
 $V(S_t) \leftarrow \text{average}(Returns(S_t))$

$V(s) = S(s) / N(s)$

Where $S(s) = G_{t_1} + G_{t_2} + \dots + G_{t_n}$

Do it incrementally

$V(s) = V(s) + \frac{1}{N(s)} (G_t - V(s))$

$V(s)_{org} = \frac{G_1 + G_2 + G_3}{3}$
 $V(s)_{update} = \frac{G_1 + G_2 + G_3 + G_4}{4} = \frac{G_1 + G_2 + G_3}{4} + \frac{G_4}{4}$
 $= \frac{G_1 + G_2 + G_3}{3} \cdot (1 - \frac{1}{4}) + \frac{G_4}{4}$
 $= V(s)_{org} + \frac{1}{4} (G_4 - V(s)_{org})$

In nonstationary Problem (i.e. Policy, reward would change)

$V(s) = V(s) + \alpha (G_t - V(s))$

- MC works when the environment is unknown
- Working with a sampled episode has a huge advantage, even when one has complete knowledge of the environment's dynamics, for example, when transition probability is complex to compute
- Cost of estimating a single state's value is independent of the total number of states. We can sample an episode starting from the state of interest, then average returns.

if: Model unavailable.

We better estimate State-action Value (i.e. $Q(s, a)$)

Challenge: Some action-state pair may never be used.

Solution: Exploring Start (try every pairs at each episode's Start)

TD: Temporary Difference Learning

MC: Estimate using actual empirical rewards

TD: Using estimated value $V(s_t)$ to estimate

$\Rightarrow$ "Bootstrapping"

$V(s_t) = V(s_t) + \alpha (r_t + \gamma V(s_{t+1}) - V(s_t))$ δ : TD error

Update a guess towards a guess

MC	TD
$V(s) = V(s) + \alpha (G_t - V(s))$	$V(s) = V(s) + \alpha (r_t + \gamma V(s_{t+1}) - V(s_t))$
Zero Bias & High Variance	Low Bias & Low Variance
Less sensitive to initial value	Sensitive to initial value
need complete sequence	incomplete sequence is also ok
need episodic environment	can use in continuous env
TD target is much lower variance than the return:	more efficient
- Return depends on many random actions, transitions, rewards - TD target depends on one random action, transition, reward	
always converge.	

$V_\pi(s) = \mathbb{E}[G_t | S_t = s]$

MC: want $V(s_t) \rightarrow G_t = r_t + \gamma r_{t+1} + \gamma^2 r_{t+2} + \dots + \gamma^n r_{t+n}$

$\therefore \mathbb{E}[V(s_t)] = \mathbb{E}[G_t | S_t = s] = V_\pi(s)$

Unbiased

TD: want $V(s_t) \rightarrow r_t + \gamma V(s_{t+1})$

(Hummus: $G_t = r_t + \gamma V(s_{t+1})$)

$\Rightarrow \mathbb{E}[V(s_{t+1})] \neq \mathbb{E}[G_t(s)]$

$\Rightarrow \mathbb{E}[V(s_{t+1})] = V_\pi(s_t)$

$\therefore$ Biased

Variance

MC: Sequence Size ↑

Variance ↑

TD: only depend on

re

Batch MC & TD

finite experience, Solution: repeat the episode k times

TD exploits Markov Property while MC not.

Markov 属性依赖

当前状态已经包含了预测未来所需的所有信息，过去的历史不再重要。

形式化地说:

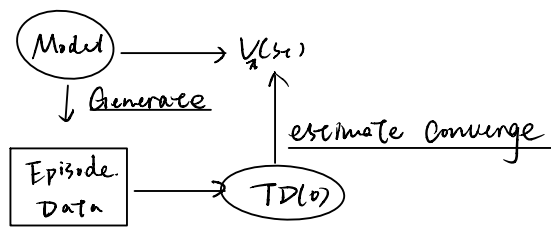
$$P(S_{t+1}, R_{t+1} | S_t, A_t) = P(S_{t+1}, R_{t+1} | S_t, A_t, \dots, S_1, A_1)$$

We Assume: $V(s_{t+1})$ contains all info from future.

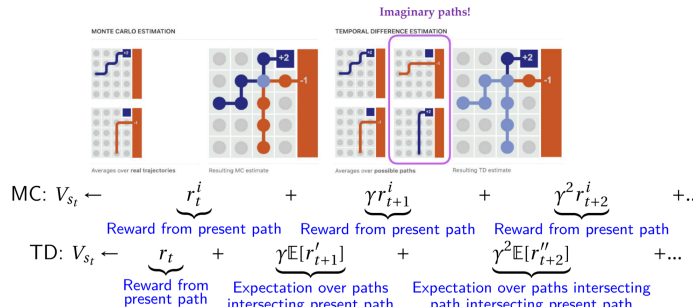
TD(0): $V(s_t) \leftarrow V(s_t) + \alpha (r_t + \gamma V(s_{t+1}) - V(s_t))$

Converges to a solution of Max Likelihood Markov Model.

Assume: There is a Markov Model



Certainty Equivalence estimate



- Bootstrapping: update involves an estimate
 - MC does not bootstrap
 - DP bootstraps
 - TD bootstraps
- Sampling: update samples an expectation
 - MC samples
 - DP does not sample
 - TD sample

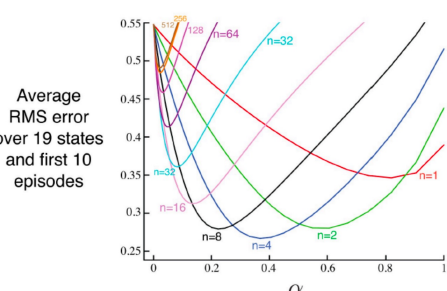
TD(λ)

"look n steps into the future"

TD(0): no look at future

TD(1): look all future $\Rightarrow$ MC

(n=∞)



Best way: Combining TD & MC

n-Step return

$$G_t^{(n)} = r_t + \gamma r_{t+1} + \dots + \gamma^{n-1} r_{t+n-1} + \gamma^n V(s_{t+n})$$

λ-return

$$G_t^\lambda = \sum_{k=0}^{\infty} (1-\lambda) \gamma^k G_{t+k}^{(n)}$$

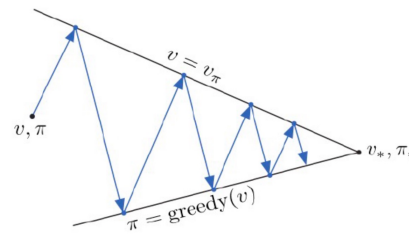
$$\sum_{k=0}^{\infty} (1-\lambda) \gamma^k = 1 \Rightarrow \text{几何级数求和}$$

Forward-view TD(λ)

$$V(s_t) \leftarrow V(s_t) + \alpha (G_t^\lambda - V(s_t)) \Rightarrow \text{Too many computation!}$$

Model-Free Control

① On-policy MC Control



- Policy evaluation estimate $V_\pi \rightarrow$ Using MC, TD, DP, ...
 - Iterative policy evaluation
- Policy improvement generate $\pi' \geq \pi$
 - Greedy policy improvement

Evaluation: MC.

Improvement: Greedy: $\pi(s) = \arg \max_a [r(s, a) + \gamma \sum_{s'} p(s'|s, a) V(s')]$
require Model(π, r)

$$\pi(s) = \arg \max_a Q(s, a)$$

Model Free

- Greedified policy meets the conditions for policy improvement:
- For all $s \in \mathcal{S}$:
$$Q_{\pi_k}(s, \pi_{k+1}(s)) = Q_{\pi_k}(s, \arg \max_a Q_{\pi_k}(s, a)) \geq Q_{\pi_k}(s, \pi_k(s))$$

- And thus must be $\pi_{k+1} \geq \pi_k$

Two assumption to obtain Convergence of MC.

1. Exploring Start

2. Policy Evaluation with infinite number of episodes.

- The exploration-exploitation dilemma!
- If we always follow the deterministic policy to collect experience, we may never have the opportunity to see and evaluate (estimate Q) of alternative actions ...
- Solutions:
 - Exploring starts: Every state-action pair has a non-zero probability of being the starting pair
 - Stochastic policies: Give up on deterministic policies and only search over stochastic or soft policies (e.g., ϵ -greedy)
 - Off-policy learning: Use a different policy to collect experience than the one you care to evaluate.

Solution 1: Exploring Start

Solution 2: ϵ -greedy

$$\pi(a|s_t) = \begin{cases} \frac{\epsilon}{|\mathcal{A}|} + 1 - \epsilon, & \text{if } a^* = \arg \max_a Q(s_t, a) \\ \frac{\epsilon}{|\mathcal{A}|}, & \text{otherwise,} \end{cases}$$

$V_{\pi^*}(s) = Q_{\pi^*}(s, \pi^*(s))$

$$= \sum_{a \in \mathcal{A}} \pi^*(a|s) Q_{\pi^*}(s, a)$$

$$= \frac{\epsilon}{|\mathcal{A}|} \sum_{a \in \mathcal{A}} Q_{\pi^*}(s, a) + (1 - \epsilon) \cdot \arg \max_a Q_{\pi^*}(s, a)$$

$$\geq \frac{\epsilon}{|\mathcal{A}|} \sum_{a \in \mathcal{A}} Q_{\pi^*}(s, a) + (1 - \epsilon) \frac{\sum_{a \in \mathcal{A}} \pi(a|s) - \frac{\epsilon}{|\mathcal{A}|}}{1 - \epsilon} Q_{\pi^*}(s, a)$$

$$= \frac{\epsilon}{|\mathcal{A}|} \sum_{a \in \mathcal{A}} Q_{\pi^*}(s, a) + \sum_{a \in \mathcal{A}} \pi(a|s) Q_{\pi^*}(s, a) - \frac{\epsilon}{|\mathcal{A}|} \sum_{a \in \mathcal{A}} Q_{\pi^*}(s, a)$$

$$= \sum_{a \in \mathcal{A}} \pi(a|s) Q_{\pi^*}(s, a) = V_{\pi^*}(s)$$

$\therefore \epsilon$ -greedy always improves policy

On-policy first-visit MC control (for ϵ -soft policies), estimates $\pi \approx \pi^*$.
Algorithm parameter: small $\epsilon > 0$
Initialize:
 $\pi \leftarrow$ an arbitrary ϵ -soft policy
 $Q(s, a) \in \mathbb{R}$ (arbitrarily), for all $s \in \mathcal{S}, a \in \mathcal{A}(s)$
 $Returns(s, a) \leftarrow$ empty list, for all $s \in \mathcal{S}, a \in \mathcal{A}(s)$
Repeat forever (for each episode):
Generate an episode following π : $S_0, A_0, R_1, \dots, S_{T-1}, A_{T-1}, R_T$
 $G \leftarrow 0$
Loop for each step of episode, $t = T-1, T-2, \dots, 0$:
 $G \leftarrow \gamma G + R_{t+1}$
Unless the pair S_t, A_t appears in $S_0, A_0, S_1, A_1, \dots, S_{t-1}, A_{t-1}$:
Append G to $Returns(S_t, A_t)$
MC $\leftarrow \frac{Q(S_t, A_t)}{Q(S_t, A_t) + \text{average}(Returns(S_t, A_t))}$ (with ties broken arbitrarily)
 ϵ -greedy $\left\{ \begin{array}{l} A^* \leftarrow \arg \max_a Q(S_t, a) \\ \text{For all } a \in \mathcal{A}(S_t): \\ \pi(a|S_t) \leftarrow \begin{cases} 1 - \epsilon + \epsilon/|\mathcal{A}(S_t)| & \text{if } a = A^* \\ \epsilon/|\mathcal{A}(S_t)| & \text{if } a \neq A^* \end{cases} \end{array} \right.$

GLIE Theorem (Greedy in the Limit with Infinite Exploration)

Definition (GLIE)

- All state-action pairs are explored infinitely many times, $\lim_{t \rightarrow \infty} N_t(s, a) = \infty$ MC guaranteed this!
- The policy converges on a greedy policy, $\lim_{t \rightarrow \infty} \pi_t(a|s) = 1$ ($a = \arg \max_{a' \in \mathcal{A}} Q_t(s, a')$)

$$\epsilon\text{-greedy } p(a^*|s_t) = \frac{\epsilon}{|\mathcal{A}|} + 1 - \epsilon \Rightarrow \text{if } \epsilon = \frac{1}{k}, \lim_{k \rightarrow \infty} p(a^*|s_t) = 1$$

Theorem (GLIE Convergence Theorem)

GLIE model-free control converges to the optimal action-value functions, $Q_* \rightarrow Q_*$

- Sample k -th episode using $\pi: [s_1, a_1, r_2, \dots, r_T] \sim \pi$
- For each state s_t and action a_t in the episode,

$$N(s_t, a_t) \leftarrow N(s_t, a_t) + 1$$

$$Q(s_t, a_t) \leftarrow Q(s_t, a_t) + \frac{1}{N(s_t, a_t)} (G_t - Q(s_t, a_t))$$

- Improve policy based on new action-value function

$$\epsilon \leftarrow \frac{1}{k}, \quad \pi \leftarrow \epsilon\text{-greedy}(Q)$$

- This GLIE Monte-Carlo control converges to the optimal action-value function, $Q(s, a) \rightarrow Q_*(s, a)$